

TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS

IFA	1	16 Jan 2026	ISSUED FOR INFORMATION	TAM (MT)	TAM (LD/MM)	ITF (IC)
IFA	0	02 Oct 2025	ISSUED FOR APPROVAL	TAM (MT)	TAM (LD/MM)	ITF (IC)
Status	Rev. Num.	Date	Description	Prepared by	Approved by	Approved by
Rev. Index						
ManaEnergy  MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage, Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco				TENDRARA PROJECT		
Contractor 			ITALFLUID BRANCH Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca	Provision of a leased train for production of LNG for Sound Energy		
Engineering 			ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) ITALY Tel:085-4224374 / fax: 085-4220742			
Facility Name TENDRARA			Location MOROCCO	Scale A4	Sheet of sheets 1/50	
Document Title HMI PHILOSOPHY				Doc. Code		
				Vendor Doc. code 21044-AU-HMI-00002		

1

Contents

2 Introduction 3

This document is property of TAM.
It will neither be shown to Third Parties not used for purposes other than those for which it has been sent.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 2 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

3	<i>HMI visualization</i>	3
3.1	<i>Screens Background</i>	3
4	<i>Texts.....</i>	4
5	<i>Process, Instrumentation And Logic-Related Lines</i>	5
6	<i>Vessels.....</i>	6
7	<i>Alarm Indications</i>	7
8	<i>Process And Logic Objects.....</i>	9
8.1	<i>Motors</i>	9
8.2	<i>On/Off Valves</i>	15
8.3	<i>Control Valves.....</i>	21
8.4	<i>Digital Signals</i>	28
8.5	<i>Analog Signals.....</i>	34
8.6	<i>PID Controllers.....</i>	42
8.7	<i>Interlocks.....</i>	49

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 3 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

2 Introduction

With the goal of developing a High-Performance visualization system, an HMI Philosophy to establish the principles of colours, looks and information to be shown and presented along the screens and process objects is a necessity.

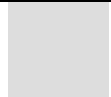
In this sense and based on good practices and guidelines studied from various sources, this document presents the standard colour coding and structure the HMI application will use for TAM158 Morocco Project.

3 HMI visualization

3.1 Screens Background

The background of screens must not cause fatigue to the eyes neither is uncomfortable to see. In this sense grey colours are the ones used the most in Standardized DCS systems.

The following colour has been selected for the background based on the ISA recommendations:

Assigned number	RGB Code	Sample
5	221,221,221	

Comparison between the options given with sample texts:

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 4 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1



4 Texts

Text colour is standardized in the unit for all related texts in general screens, block icons and faceplates. Please refer to the following table for more details:

Numeration	RGB code	Sample	Description
1	0,0,0		Non-changing texts (Example: Descriptions, tag names, etc).
2	0,0,255		Dynamic values and states. For analogue values, denotes read-only values.
3	0,128,1		Analogue values that can be adjusted by operator.
4	255,0,255		Denotes an equipment is in Simulation mode.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 5 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

5 Process, Instrumentation And Logic-Related Lines

Process lines will not change depending on the liquid/gas they contain. They will have a maximum of three thickness levels. A Process line will always be a solid and thicker line compared to instrumentation lines and logic-related lines.

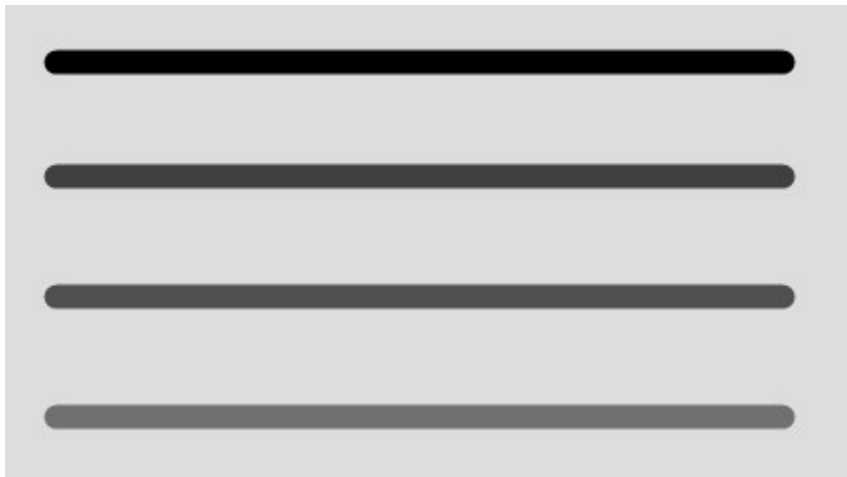
Instrumentation and Logic related lines should have the lowest thickness of all the lines. They can use the same or a darker colour compared to the Process lines.

Logic-related lines will be dashed or dashed-dotted.

Taking into consideration the background colour chosen for the project, the following options are proposed for this project:

Assigned number	RGB Code	Sample
1	0,0,0	
2	64,64,64	
3	80,80,80	
4	112,112,112	

All the options presented above are presented below for comparison purposes against the background colour proposed in this document

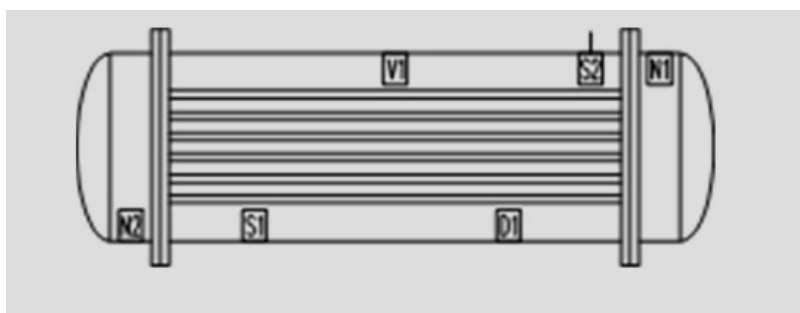


	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 6 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

6 Vessels

For simplification purposes, all Chillers, Compressors, Vessels, Economizers, etc. Are called Vessels in this document. They will be 2-D objects with solid and uniform body colour that will be the same as the one chosen for the background. Border line will be one pixel thick and solid black colour RGB (0,0,0).

Below there's a example of a vessel:



	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 7 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

7 Alarm Indications

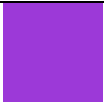
There are going to be 4 types or Alarms:

- Trip: It has priority 1. A Trip condition is related with abnormal situation that needs to be avoided. It triggers an Abnormal Shutdown for the system it belongs to. Example: High-High Pressure in a tank, ESD Button pressed, Low-Low Oil Level detected by a Level switch, etc.
- Interlock: It has priority 2. It's a status that inhibits or disables certain actions or controls in the system. It will not directly trigger an abnormal shutdown though interferes in the normal operation of the system.
- Warning: It has priority 3. Indicates when there's a situation outside of the normal conditions of plan operation though by itself doesn't trigger a shutdown. Example: High Pressure in a tank (above normal though below trip setpoint), Low Oil Level detected by a Level Switch.
- Diagnostic: It has priority 4. A Failure condition is related with an equipment that is not healthy or is not behaving in the way it is supposed to. Example: A Motor failed to start, Pressure transmitter's wire broken, Control Valve Mismatching Command vs Feedback, Communication with another system has been interrupted, etc. When the source of the diagnostic message is important for the system it belongs to, a separate Trip Signal is generated.

The colours used for the alarm view and to represent the alarms on the screens are presented below (For Diagnostic alarms, there are three options presented. Option 1 is the one used for block icon and faceplates examples later in this document):

Alarm Type	RGB Code	Sample	
Trip	255,0,0		
Interlock	255,102,0		
Warning	255,255,0		
Diagnostic (Option 1)	18,142,255		

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 8 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Diagnostic (Option 2)	255,0,255		
Diagnostic (Option 3)	156,57,216		

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 9 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

8 Process And Logic Objects

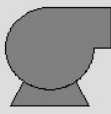
8.1 Motors

Block Icons to be used in the various process screens are going to be standard, 2-D representations with no deep details in the non-changing body of the object.

Fans, blowers, mixers, etc will not have animation neither blinking related to movement and/or running status.

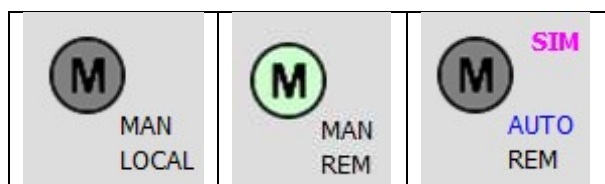
If there's going to be any equipment whose running/stopped status is not received by our system, the body colour of that equipment will be the same as the background.

Please find below the options presented for different statuses for the block icon:

Status	Icon Example	RGB Code
Stopped		(128,128,128)
Running		(204,255,204)

Block Icon

The block icon of a motor will vary depending on whether it's representing a pump, fan, general purpose motor, etc. For simplicity, a block icon includes just the icon showing stopped/running status by its colours and text indicators about its status. The order of the text statuses is the same for all motors from top to bottom: Simulation indicator, Auto/Manual indicator and Local/Remote status (if applicable) are presented. Tag name will be dynamically showed/hidden by a button for all the equipment.



Simulation Mode indicator when visible is in BOLD font with colour RGB(255,0,255). Automatic mode text indicator is the only one using a different font colour RGB(0,0,255) to be easier and faster for

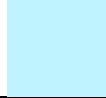
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 10 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

operator to look all the equipment on screen and recognize which equipment are in the normal/desired status.

Faceplates

The faceplate is standard for all types of motors and contains several tabs with different features and indications useful for operation. For all the tabs, always the tag name and type are shown on the header.

Colour coding is applied also to the faceplate, for buttons and indicators as follows:

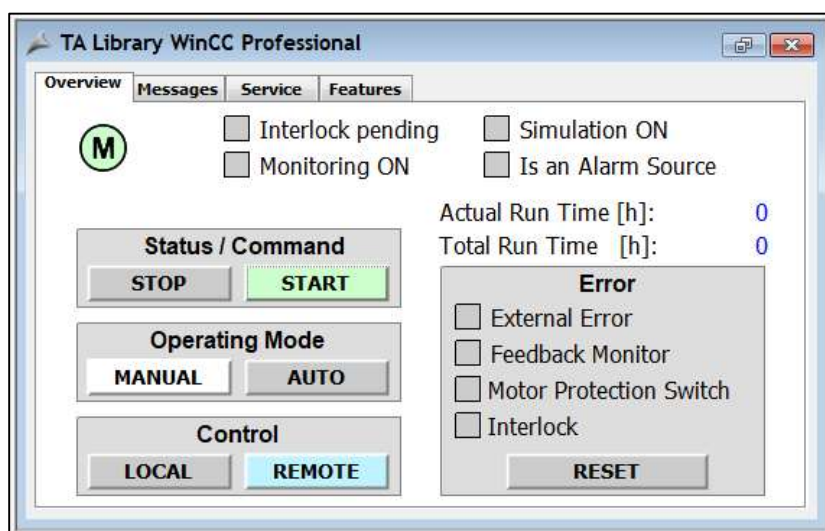
Status	RGB Code	Sample
Deactivated Status	204,204,204	
Active 1	190,243,255	
Active 2	255,255,255	
Error indication	255,170,170	

Active 1 is used when the normal/ideal mode is active (Like Automatic Mode, Monitoring ON, etc).
Active 2 is used for the rest of statuses.

Simulation mode follows its colour coding, the same goes for stopped and running status.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 11 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Overview Tab

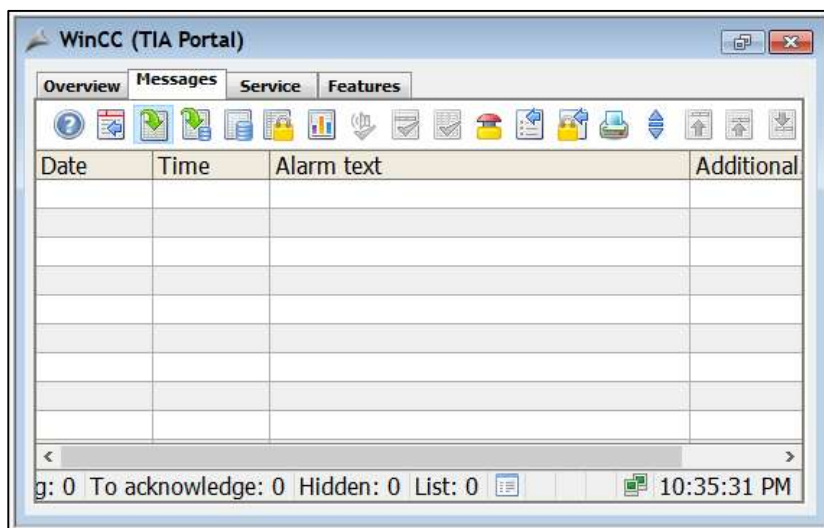


On the Overview tab, several statuses and commands are presented:

- 1) "Interlock pending" indicates Interlock signal is being received.
- 2) "Monitoring ON" indicator lights up when Feedback monitoring is enabled.
- 3) "Simulation ON" is lighted-up when motor simulation is activated.
- 4) Indication if any error from the respective motor is an Alarm or Trip source for the whole system.
- 5) Running hours of the current run and the total running hours of the motor are presented in this area.
- 6) Manual Start/Stop commands from HMI are presented here. A motor must be in remote mode and manual mode to be able to be commanded by these buttons.
- 7) Buttons to switch between Manual and Auto Mode are presented here. Button background colour lights up when the respective mode is selected.
- 8) Buttons to switch between Local and Remote Mode are presented here. Button background colour lights up when the respective mode is selected. This area is invisible when Motor doesn't have Local Mode.
- 9) Any of the statuses here trigger the general error status of the motor. "External Error" is connected to the hardwired input Motor Fault. "Feedback Error" lights-up when there's no feedback received or there's a mismatch in the feedback. "Motor protection switch" error is connected to the respective Block input, "Interlock" error is connected to the respective Block input.

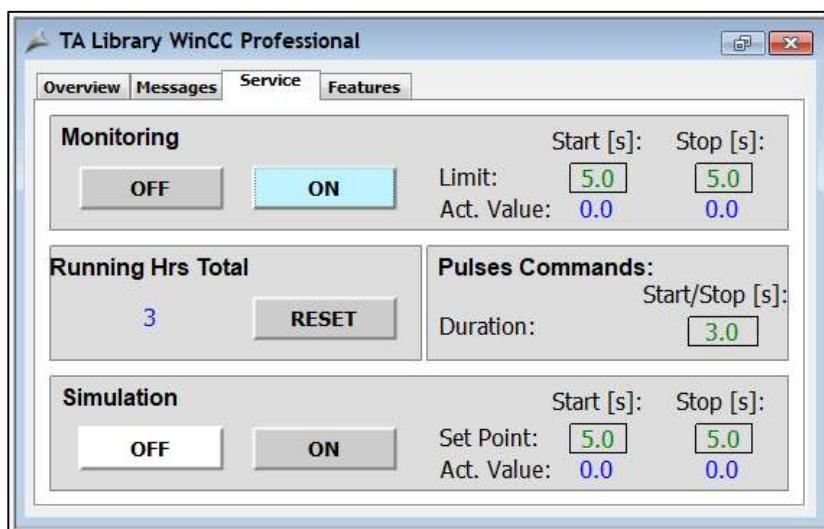
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 12 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Message Tab



On “Messages” tab, user can find an alarm window that displays only the messages related to the motor who the faceplate belongs to. Time when the message was triggered, status of the alarm and description are displayed. The Background coding used for the general Alarm screen is applied as well for specific Motor alarm messages.

Service Tab



On “Service” tab, several statuses and features are available:

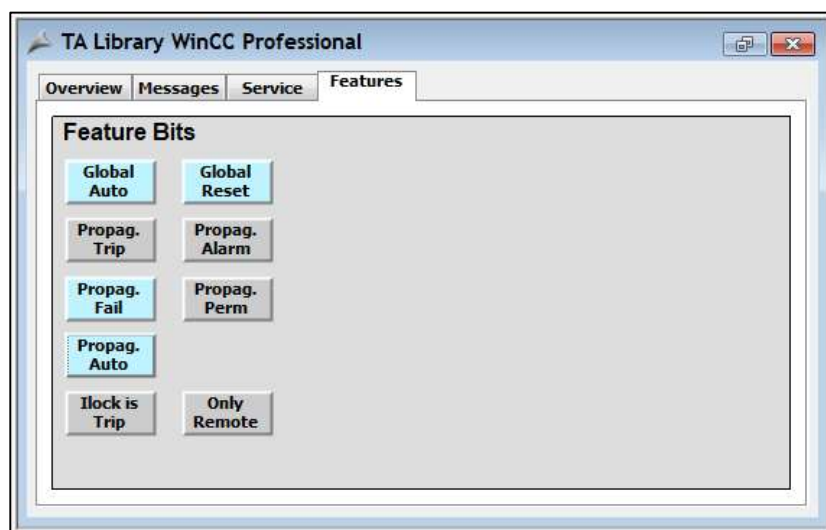
- 1) Monitoring settings. Monitoring feedback feature can be enabled/disabled using the respective buttons. Time limits that check the time elapsed from the moment a

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 13 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

command is issued and the maximum time the correct feedback must be received before triggering an alarm/trip are presented here. There's an individual limit for start and stop respectively. The actual time elapsed after command was issued and before feedback is received are presented as well.

- 2) Total Running time of the motor is shown here. Reset button will set to zero this value when pressed.
- 3) Setpoint for duration of start/stop pulse commands. If the motor is using start/stop command as pulses and not as continuous commands, then this parameter is important.
- 4) Simulation Mode settings. Simulation mode can be activated here. In Simulation mode, no commands to the real motor are sent neither feedback is monitored. This mode **MUST BE USED ONLY** during commissioning and/or troubleshooting and not during normal operation. The time setpoints presented here are the ones the control block of the motor will use to simulate start/stop feedback after a start/stop command has been issued. Elapsed times for these timers are presented here as well.

Features Tab



These are the main features that TA Library has. Based on the requirements of every project, these features are enabled/disabled. Light blue background means is enabled, Gray background means the opposite. A motor has the following features:

- **Global Auto:** When enabled, Motor will accept global requests sent from the Compressor main control block to switch to Auto mode all the equipment of the Compressor system. When disabled, switching to auto mode can be done by motor faceplate exclusively.
- **Global Reset:** When enabled, Motor will accept global requests sent from the Compressor main control block to reset all Alarm, Trips and Failures of the

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 14 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

equipment of the Compressor system. When disabled, reset commands can be done by motor faceplate exclusively.

- **Propagate Trip:** When enabled, if Motor has been configured as a Trip source, any error status of motor (except feedback error) will send a Global Trip to the Compressor system, tripping the whole package. When disabled, only this motor will stop and be in Error status if any error occurs (External Fault, Motor Protection Switch, Interlock) except for Feedback Error.
- **Propagate Alarm:** When enabled, if the Motor has been configured as an Alarm source, any Alarm status of motor will send a Global Alarm to the Compressor system, though will not trip the whole package, just will stop this motor. When disabled, no alarm status will be sent to Compressor main control block, only this motor will stop and be in Error status if any error occurs (Feedback, External Fault, Motor Protection Switch, Interlock).
- **Propagate Failure:** When enabled, if Motor has been configured as a Trip source, any feedback failure of the motor will send a Global Failure to the Compressor system, tripping the whole package. When disabled, only this motor will stop and be in Error status if any feedback error occurs.
- **Propagate Permissive:** When enabled, any error present in the motor (Feedback error, External Fault, Motor Protection Switch, Interlock) will send a No-Ready status to the Compressor system and Compressor package will not be able to Start until the error is cleared. When disabled, only this motor will be in Error status and will not interfere with the Ready to Start status of the Compressor Package.
- **Propagate Auto Mode:** When enabled, if the motor is in Manual Mode, it will send a No-Ready status to the Compressor system and Compressor package will not be able to Start until it is switched to Auto Mode and a Reset Alarm command is executed. When disabled, Manual/Auto mode will not interfere with the Ready to Start status of the Compressor Package.
- **Interlock is Trip:** When enabled, Interlock error generates a Trip to this motor and depending on other feature bits, can be a Trip trigger for the whole Compressor package. When disabled, Interlock error is just a permissive for the motor to start, when running will not stop the motor.
- **Only Remote:** When enabled, motor only has remote mode (controlled by PLC and commands are generated via HMI) and respective Loca/Remote indications and switch buttons are hidden from faceplates and block icons on screen. When disabled, Local/Remote statuses and switch buttons are presented on faceplates and block icons on screen.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 15 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

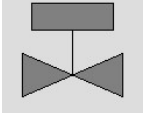
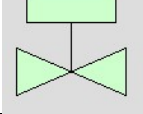
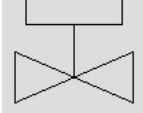
8.2 On/Off Valves

Block Icons to be used in the various process screens are going to be standard, 2-D representations with no deep details in the non-changing body of the object.

Valves will not have animation neither blinking related to movement and/or opened/closed status.

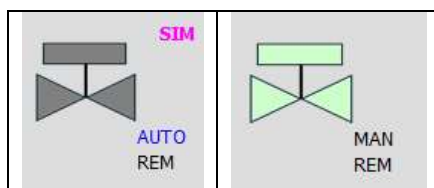
If there's going to be any equipment whose opened/closed status is not received by our system, the body colour of that equipment will be the same as the background.

Please find below the options presented for different statuses for the block icon:

Status	Icon Example	RGB Code
Closed		(128,128,128)
Opened		(204,255,204)
No Status being received at the Control system		Hollow, same as background

Block Icon

The block icon of a On/Off Valve is standard for all the valves in the project. For simplicity, a block icon includes just the icon showing Closed/Opened status by its colours and text indicators about its status. The order of the text statuses is the same for all valves from top to bottom: Simulation indicator, Auto/Manual indicator and Local/Remote status (if applicable) are presented. Tag name will be dynamically showed/hidden by a button for all the equipment.



Simulation Mode indicator when visible is in BOLD font with colour RGB(255,0,255). Automatic mode text indicator is the only one using a different font colour RGB(0,0,255) to be easier and

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 16 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

faster for operator to look all the equipment on screen and recognize which equipment are in the normal/desired status.

Faceplates

The faceplate is standard for all On/off valves and contains several tabs with different features and indications useful for operation. For all the tabs, always the tag name and type are shown on the header.

Colour coding is applied also to the faceplate, for buttons and indicators as follows:

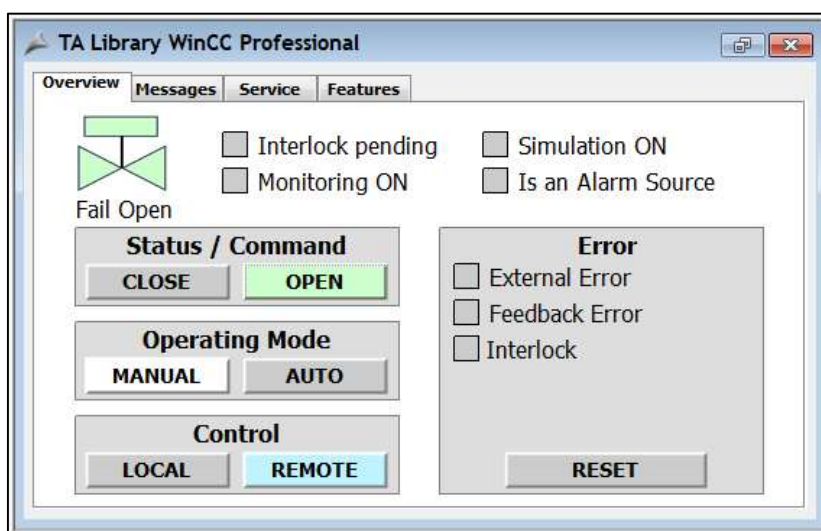
Status	RGB Code	Sample
Deactivated Status	204,204,204	
Active 1	190,243,255	
Active 2	255,255,255	
Error indication	255,170,170	

Active 1 is used when the normal/ideal mode is active (Like Automatic Mode, Monitoring ON, etc). Active 2 is used for the rest of statuses.

Simulation mode follows its colour coding, the same goes for opened and closed status.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 17 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Overview Tab

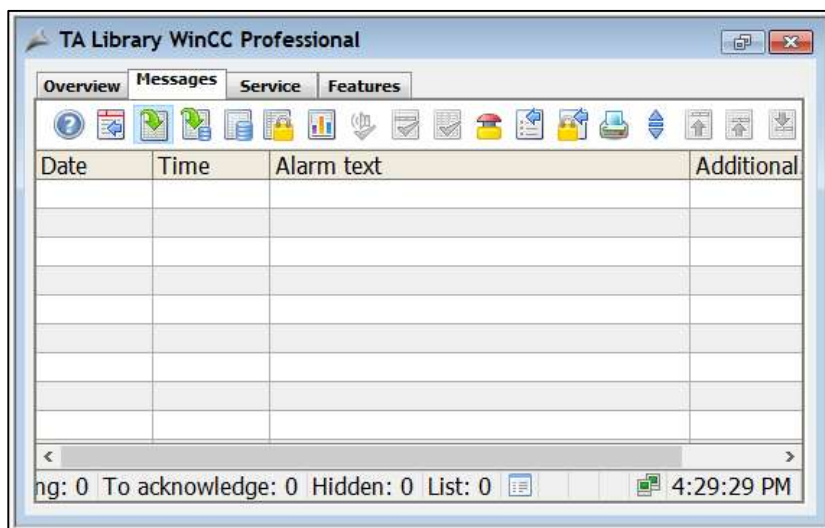


On the Overview tab, several statuses are presented:

- 1) Valve type (fail close, fail open)
- 2) "Interlock pending" that indicates Interlock signal is being received.
- 3) "Monitoring ON" indicator lights up in blue when Feedback monitoring is enabled.
- 4) "Simulation ON" is lighted-up when Valve simulation is activated.
- 5) Indicator about what generates a valve error towards the system. "Is a Trip source/Is an Alarm source" text is shown.
- 6) Manual commands to Open and Close the Valve are presented here as buttons.
- 7) Buttons to switch between Manual and Auto Mode.
- 8) Buttons to switch between Local and Remote Mode. In this project, solenoid valves only have Remote mode.
- 9) Any of the statuses here trigger the general error status of the Valve. Because there are no hardwired position readbacks in this project nor external fault signals, "External Error", "Feedback Error" neither "Interlock" is used and will not generate any valve-related error.

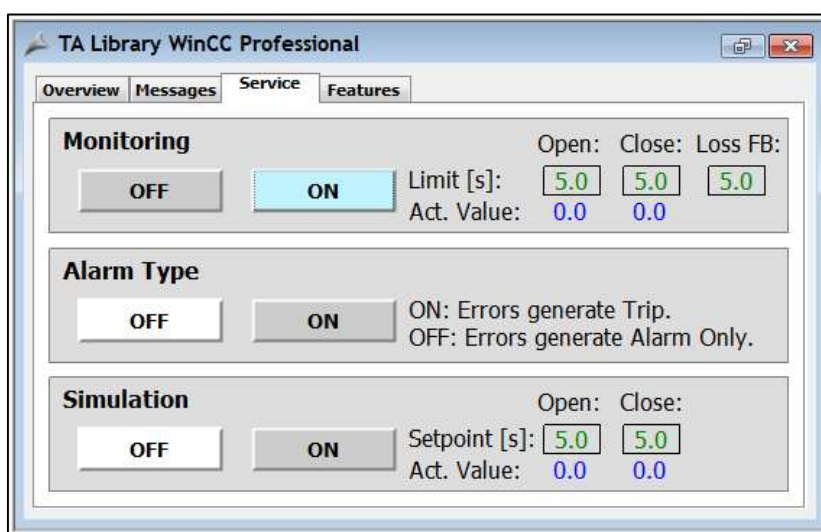
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 18 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Message Tab



On “Messages” tab, user can find an alarm window that displays only the messages related to the Valve who the faceplate belongs to. Time when the message was triggered, status of the alarm and description are displayed. The Background coding used for the general Alarm screen is applied as well for specific alarm messages.

Service Tab



- 1) Monitoring settings. Monitoring feedback feature can be enabled/disabled using the respective buttons. Time limits that check the time elapsed from the moment a command is issued and the maximum time the correct feedback must be received

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 19 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

before triggering an alarm/trip are presented here. There's an individual limit for open and close respectively. The actual time elapsed after command was issued and before feedback is received are presented as well.

- 2) Alarm Type buttons to switch between Fail/Trip generation and Alarm respectively.
- 3) Simulation Mode settings. Simulation mode can be activated here. In Simulation mode, no commands to the real valve are sent neither feedback is monitored. This mode **MUST BE USED ONLY** during commissioning and/or troubleshooting and not during normal operation. The time setpoints presented here are the ones the control block of the valve will use to simulate open/close feedback after an open/close command has been issued. Elapsed times for these timers are presented here as well.

Features Tab



These are the main features that TA Library has. Based on the requirements of every project, these features are enabled/disabled. Light blue background means is enabled, grey background means the opposite. A Valve has the following features:

- 1) Global Auto: When enabled, Valve will accept global requests sent from the System main control block to switch to Auto mode all the equipment of the system. When disabled, switching to auto mode can be done by Valve faceplate exclusively.
- 2) Global Reset: When enabled, Valve will accept global requests sent from the System main control block to reset all Alarm, Trips and Failures of the equipment of the system. When disabled, reset commands can be done by valve faceplate exclusively.
- 3) Propagate Trip: When enabled, if Valve has been configured as a Trip source, any error status of valve (except feedback error) will send a Global Trip to the system, tripping the whole package. When disabled, only this valve will go to fail position and be in Error status if any error occurs (External Fault, Interlock) except for Feedback Error.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 20 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

- 4) Propagate Alarm: When enabled, if the Valve has been configured as an Alarm source, any Alarm status of valve will send a Global Alarm to the system, though will not trip the whole package, just will send this valve to fail position. When disabled, no alarm status will be sent to System main control block, only this valve will go to fail position and be in Error status if any error occurs (Feedback, External Fault, Interlock).
- 5) Propagate Failure: When enabled, if Valve has been configured as a Trip source, any feedback failure of the Valve will send a Global Failure to the system, tripping the whole package. When disabled, only this Valve will go to fail position and be in Error status if any feedback error occurs.
- 6) Propagate Permissive: When enabled, any error present in the Valve (Feedback error, External Fault, Interlock) will send a No-Ready status to the System and Compressor package will not be able to Start until the error is cleared. When disabled, only this Valve will be in Error status and will not interfere with the Ready to Start status of the System Package.
- 7) Propagate Auto Mode: When enabled, if the Valve is in Manual Mode, it will send a No-Ready status to the System and the package will not be able to Start until it is switched to Auto Mode and a Reset Alarm command is executed. When disabled, Manual/Auto mode will not interfere with the Ready to Start status of the System Package.
- 8) Fail Type of the Valve: By text the actual settings are shown on the body of the button. When the Valve is set as Fail Open Valve, the body of the button will be light blue and that text will be displayed. When it is set as Fail Close Valve, the body of the button will be Gray and that text will be displayed.
- 9) Only Remote: When enabled, Valve only has remote mode (controlled by PLC and commands are generated via HMI) and respective Loca/Remote indications and switch buttons are hidden from faceplates and block icons on screen. When disabled, Local/Remote statuses and switch buttons are presented on faceplates and block icons on screen.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 21 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

8.3 Control Valves

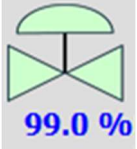
Block Icons to be used in the various process screens are going to be standard, 2-D representations with no deep details in the non-changing body of the object.

Control Valves will not have animation neither blinking related to movement and/or opened/closed status.

Opening percentage is presented in blue colour font RGB (0,0,255).

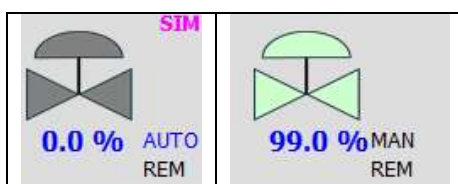
If there's going to be any equipment whose position is not received by our system, the body colour of that equipment will be the same as the background.

Please find below the options presented for different statuses for the block icon:

Status	Icon Example	RGB Code
Closed		(128,128,128)
Opened		(204,255,204)

Block Icon

The block icon of a Control Valve is standard for all the valves in the project. For simplicity, a block icon includes just the icon showing Closed/Opened status by its colours, opening in percentage and text indicators about its status. The order of the text statuses is the same for all valves from top to bottom: Simulation indicator, Auto/Manual indicator and Local/Remote status (if applicable) are presented. Tag name will be dynamically showed/hidden by a button for all the equipment.



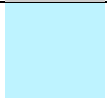
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 22 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Simulation Mode indicator when visible is in BOLD font with colour RGB(255,0,255). Automatic mode text indicator is the only one using a different font colour RGB(0,0,255) to be easier and faster for operator to look all the equipment on screen and recognize which equipment are in the normal/desired status.

Faceplates

The faceplate is standard for all Control valves and contains several tabs with different features and indications useful for operation. For all the tabs, always the tag name and type are shown on the header.

Colour coding is applied also to the faceplate, for buttons and indicators as follows:

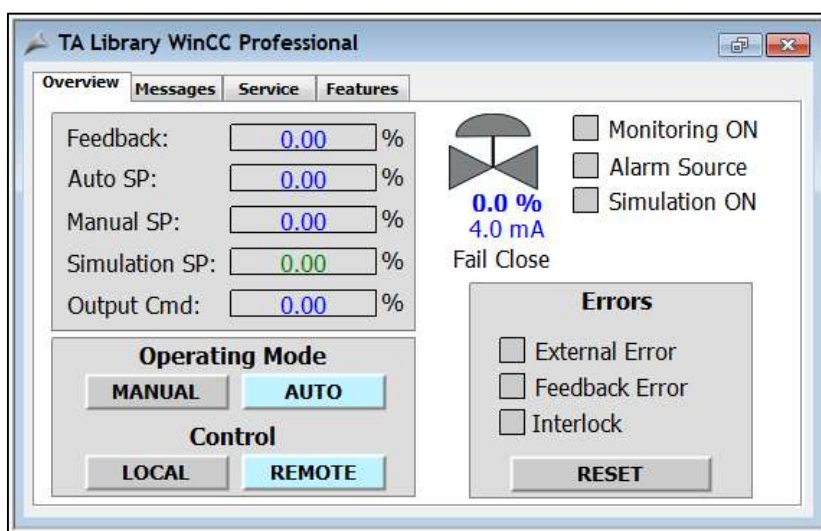
Status	RGB Code	Sample
Deactivated Status	204,204,204	
Active 1	190,243,255	
Active 2	255,255,255	
Error indication	255,170,170	

Active 1 is used when the normal/ideal mode is active (Like Automatic Mode, Monitoring ON, etc). Active 2 is used for the rest of statuses.

Simulation mode follows its colour coding, the same goes for closed and opened status.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 23 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Overview Tab



On the Overview tab, several statuses are presented as:

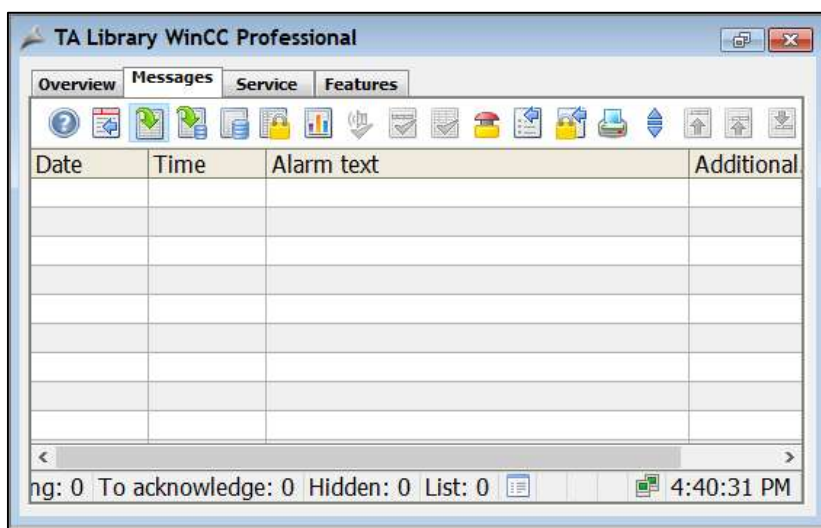
- 1) In this area we found several values related to the valve:
 - “Feedback” is the value read from the positioner by Analog Input module if Feedback Monitoring is enabled, otherwise it's equal to the command sent to the valve by the Control Valve program Block.
 - “Auto Setpoint” is the position command sent by PLC controller's Logic to the Valve.
 - “Manual Setpoint” is the position command sent by operator using this faceplate when Valve is in Manual Mode.
 - “Simulation Setpoint” is the simulated position the Valve will reflect in its Control Block for the PLC logic when the valve is in Simulation Mode (Please note Simulation Mode should be used ONLY during commissioning and/or troubleshooting and under supervision by TA personnel).
 - Output Command is the command in percentage sent to the Valve, regardless of if the valve is in auto/manual mode and simulation/process mode.
- 2) Valve is graphically represented here. Valve type (Fail Open/Close) and calculated current supplied to the valve via the Analog Outputs channel are presented as well. Ramp status indication is presented here too if it is enabled, otherwise is hidden.
- 3) Several Statuses are shown on this area:
 - Monitoring ON: If Monitoring of feedback is enabled, the square indicator lights up.
 - Error is Alarm/Trip indicator is shown respectively according to the configuration of the valve as an alarm/trip source for the whole System.
 - Simulation ON: When Simulation Mode is active, the square indicator lights up.
- 4) Command buttons presented here are:
 - Manual, Auto Mode: Buttons to switch between Manual and Automatic Mode.
 - Local, Remote Mode: Buttons to switch between Local and Remote Mode.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 24 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

5) Indicators if Valve is in any error status. They are:

- External Error, Indicator lights up if Valve control block's input for external fault supplies that signal.
- Feedback Error, Indicator lights up if with "Monitoring ON", Valve control block detects Maximum allowed deviation between Command and Feedback position has been surpassed.
- Interlock, Indicator lights up if Valve control block's input for Interlock supplies that signal.
- Reset Button, for any of the three mentioned errors.

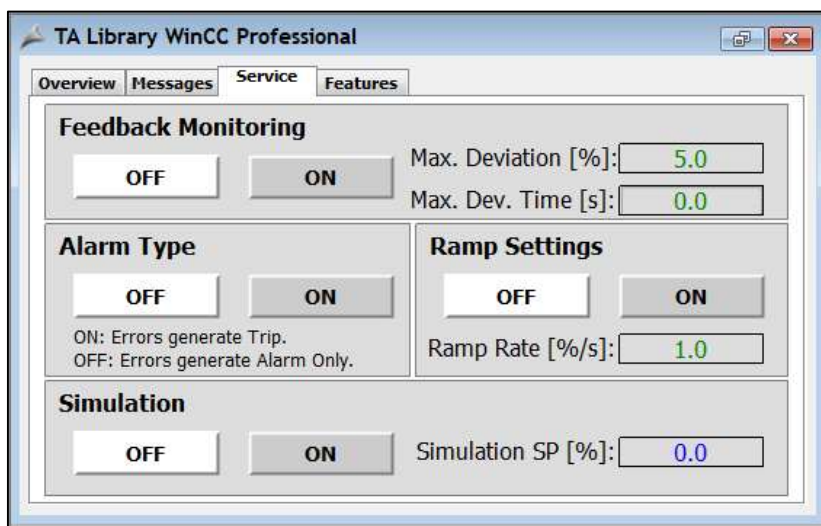
Message Tab



On "Messages" tab, user can find an alarm window that displays only the messages related to the Valve who the faceplate belongs to. Time when the message was triggered, status of the alarm and description are displayed. The Background coding used for the general Alarm screen is applied as well for specific alarm messages.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 25 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Service Tab

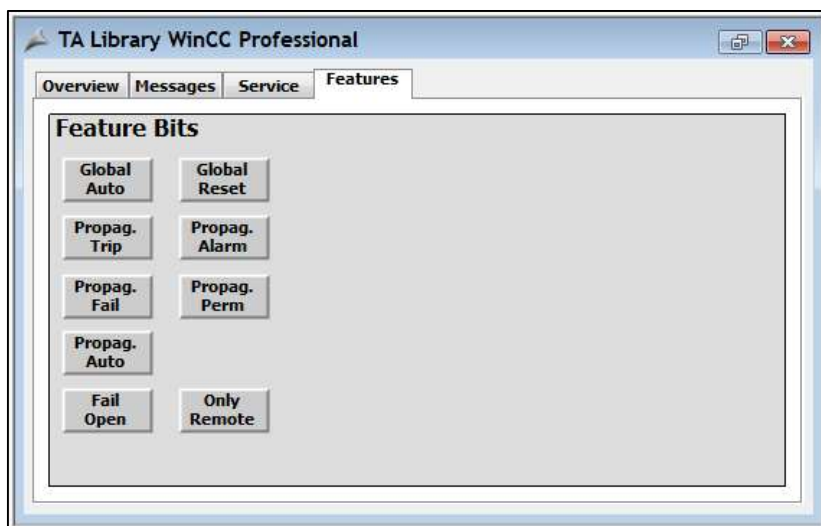


On "Service" tab, several statuses and features are presented as:

- 1) Monitoring settings. Monitoring feedback feature can be enabled/disabled using the respective buttons. "Maximum Deviation" is the maximum difference allowed between command to the valve and feedback received. "Max. Deviation time" is the delay time before an alarm/trip is generated when Command and Feedback difference is greater or equal than Maximum Deviation limit.
- 2) Alarm Type buttons to switch between Fail/Trip and Alarm generation respectively. When any error is detected in the Valve (Feedback Error, External Error, Interlock) depending on this feature, the Valve will generate a Trip/Fail or alarm only. Whether this Alarm/Trip/Fault is passed to the Whole system depends on the Feature Bits.
- 3) Ramp Feature: With this setting On, Commands in Automatic Mode (Change in Auto Setpoint) are applied progressively Up/Down at the rate presented in this area as well.
 - On/Off Buttons to Enable/Disable Ramp feature.
 - Ramp Rate: in percentage per second, is the rate of change Valve Control Block will use to ramp up/down the command sent to the valve to get to the target "Auto Setpoint".
- 4) Simulation Mode settings. Simulation mode can be activated here. In Simulation mode, no commands to the real valve are sent neither feedback is monitored. This mode **MUST BE USED ONLY** during commissioning and/or troubleshooting and not during normal operation. Simulation Setpoint is the position that Valve Control Block will use in Simulation Mode for the control logic.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 26 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Feature Tab



4) These are the main features that TA Library has. Based on the requirements of every project, these features are enabled/disabled. Light blue background means is enabled, grey background means the opposite. A solenoid valve has the following features:

- **Global Auto:** When enabled, Valve will accept global requests sent from the System main control block to switch to Auto mode all the equipment of the System. When disabled, switching to auto mode can be done by Valve faceplate exclusively.
- **Global Reset:** When enabled, Valve will accept global requests sent from the System main control block to reset all Alarm, Trips and Failures of the equipment of the system. When disabled, reset commands can be done by valve faceplate exclusively.
- **Propagate Trip:** When enabled, if Valve has been configured as a Trip source, any error status of valve (except feedback error) will send a Global Trip to the System, tripping the whole package. When disabled, only this valve will go to fail position and be in Error status if any error occurs (External Fault, Interlock) except for Feedback Error.
- **Propagate Alarm:** When enabled, if the Valve has been configured as an Alarm source, any Alarm status of valve will send a Global Alarm to the Compressor system, though will not trip the whole package, just will send this valve to fail position. When disabled, no alarm status will be sent to System main control block, only this valve will go to fail position and be in Error status if any error occurs (Feedback, External Fault, Interlock).
- **Propagate Failure:** When enabled, if Valve has been configured as a Trip source, any feedback failure of the Valve will send a Global Failure to the System, tripping the whole package. When disabled, only this Valve will go to fail position and be in Error status if any feedback error occurs.
- **Propagate Permissive:** When enabled, any error present in the Valve (Feedback error, External Fault, Interlock) will send a No-Ready status to the Whole system

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 27 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

and package will not be able to Start until the error is cleared. When disabled, only this Valve will be in Error status and will not interfere with the Ready to Start status of the whole Package.

- Propagate Auto Mode: When enabled, if the Valve is in Manual Mode, it will send a No-Ready status to the System and the package will not be able to Start until it is switched to Auto Mode and a Reset Alarm command is executed. When disabled, Manual/Auto mode will not interfere with the Ready to Start status of the whole Package.
- Fail Type of the Valve: By text the actual setting is shown on the body of the button. When the Valve is set as Fail Open Valve, the body of the button will be Light blue and that text will be displayed. When it is set as Fail Close Valve, the body of the button will be Grey and that text will be displayed.
- Only Remote: When enabled, Valve only has remote mode (controlled by PLC and commands are generated via HMI) and respective Loca/Remote indications and switch buttons are hidden from faceplates and block icons on screen. When disabled, Local/Remote statuses and switch buttons are presented on faceplates and block icons on screen.

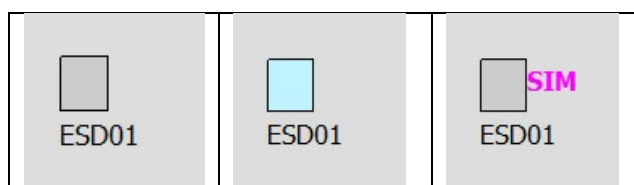
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 28 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

8.4 Digital Signals

Crucial Digital signals will be presented on the system in Block icons and faceplates as well.

BLOCK ICON

The block icon of a Digital Signal is standard in the project. For simplicity, a block icon includes just the icon showing Off/On status by colour and/or Alarm/Trip status if the signal was configured as an Alarm/Trip for the whole system; a status text indicating when Signal is in Simulation mode and Tag name that will be dynamically showed/hidden by a button for all the equipment.

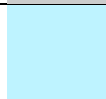
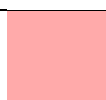


Simulation Mode indicator when visible is in BOLD font with colour RGB(255,0,255). Body of the rectangular status indicator will follow yellow RGB colour for Alarm status if the digital signal was configured as an alarm source, the same applies for Red RGB Colour when was configured as a Trip source.

FACEPLATES

The faceplate is standard for all Digital Signals and contains several tabs with different features and indications useful for operation. For all the tabs, always the tag name and type are shown on the header.

Colour coding is applied also to the faceplate, for buttons and indicators as follows:

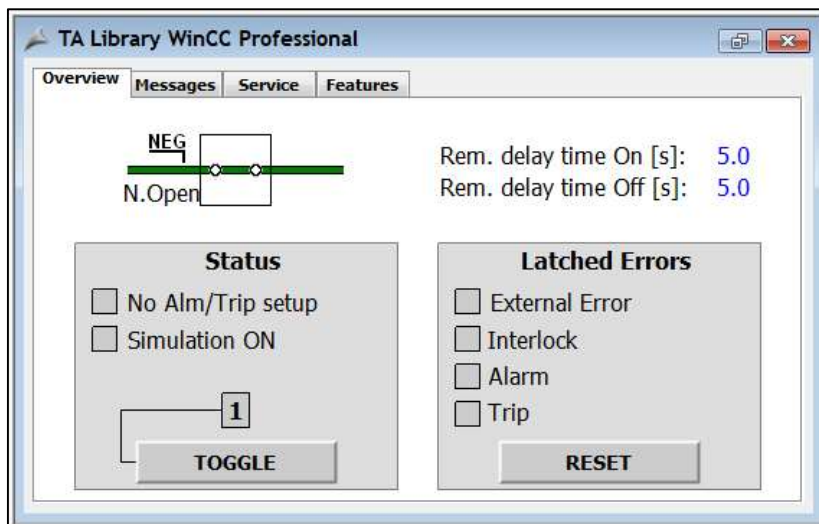
Status	RGB Code	Sample
Deactivated Status	204,204,204	
Active 1	190,243,255	
Active 2	255,255,255	
Error indication	255,170,170	

Active 1 is used when the normal/ideal mode is active (Like No Alm/Trip setup, etc). Active 2 is used for the rest of statuses.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 29 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Simulation mode follows its colour coding, the same goes for ON, OFF.

Overview Tab



Digital signal faceplate has one unique tab, the information and features contained on it are:

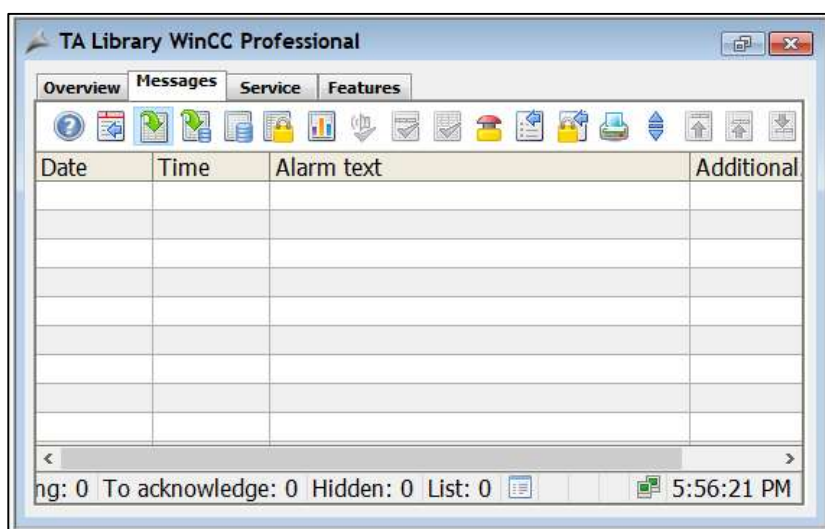
- 1) Graphic representation of the Digital signal as a contact. An acronym indicating whether is Normally Closed/Normally Open is shown as well.
- 2) All the statuses a Digital Input has are presented here:
 - Alarm/Trip Enabled: Lights up and shows the corresponding status when Digital Input is configured to generate an Alarm or Trip when is not in its normal position (Example: Is open when is set as normally closed). When Digital signal is not configured as an alarm neither trip source, "No Alm/Trip setup" text is displayed.
 - Simulation ON: Lights up when Simulation Mode is active.
 - Simulated Value: The simulated value of the Digital Input is shown as text. A Toggle button for the Simulated value is presented as well.
- 3) There's a delay timer when a signal transitions from ON to OFF and vice versa. The time setpoints are in the Service Tab though the remaining time of those delay timers are presented here.
- 4) All the statuses a Digital Input has are presented here:
 - External Error: Lights up when any external error that is related to the Digital Input occurs. It's detected by the Digital Input Control Block.
 - Interlock: Lights up if Digital Input control block's input for Interlock supplies that signal.
 - Alarm: Lights up when Digital Input is in the opposite status as its normal position (Example: It's open when normally is closed) and Digital Input was configured as an Alarm source.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 30 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

- Trip: Lights up when Digital Input is in the opposite status as its normal position and Digital Input was configured as an Alarm source.
- Reset Button for any error status related to the Digital Input.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 31 / 50	
		Contractor ITALFLUID MOROCCO BRANCH	Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Message Tab



On “Messages” tab, user can find an alarm window that displays only the messages related to the Digital Signal who the faceplate belongs to. Time when the message was triggered, status of the alarm and description are displayed. The Background coding used for the general Alarm screen is applied as well for specific alarm messages.

Service Tab



On “Service” tab, several statuses and features are available:

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 32 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

- 1) Delay settings. As mentioned before, when a Digital Input transitions from open to close and vice versa, there's a delay timer for each transition that allows Control program to be more flexible and take certain actions of each transition if required. Time Setpoints for when transitions from Off->On and On->Off are presented respectively. The actual time elapsed of an ongoing transition is shown below.
- 2) Buttons to Enable/Disable Digital Input as a Trip Source when it is not in its normally open/close status.
- 3) Buttons to Enable/Disable Digital Input as an Alarm Source when it is not in its normally open/close status.
- 4) Simulation Mode settings. Simulation mode can be activated here. In Simulation mode, no status is read from the physical Digital Input nor monitored. The value/status used for the control logic is the one simulated and modified by the toggle button on Overview Tab. This mode **MUST BE USED ONLY** during commissioning and/or troubleshooting **UNDER TA PERSONNEL APPROVAL** and not during normal operation.

Features Tab



These are the main features that TA Library has. Based on the requirements of every project, these features are enabled/disabled. Light blue background means is enabled, Grey background means the opposite. A Digital Signal has the following features:

- **Global Reset:** When enabled, Digital Input will accept global requests sent from the System main control block to reset all Alarm, Trips and Failures of the equipment of the system. When disabled, reset commands can be done by Digital Input faceplate exclusively.
- **Propagate Trip:** When enabled, if Digital Input has been configured as a Trip source, any error status or if it is not in its normal open/close position, Digital Input control block will send a Global Trip to the System, tripping the whole package. When disabled, only this Digital Input will be in Error status if any error occurs (External Fault, Interlock) or when is not in its normal open/close position.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 33 / 50	
	 Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
 Engineering ITALFLUID Geoenergy S.r.l.		Document Title HMI PHILOSOPHY	16/01/2026	1

- **Propagate Alarm:** When enabled, if Digital Input has been configured as an Alarm source, any error status or if it is not in its normal open/close position, Digital Input control block will send a Global Alarm to the System. When disabled, only this Digital Input will be in Alarm status if any error occurs (External Fault, Interlock) or when is not in its normal open/close position.
- **Propagate Permissive:** When enabled, any error present in the Digital Input (External Fault, Interlock) or if it is not in its normal open/close position, will send a No-Ready status to the System and the package will not be able to Start until the error is cleared. When disabled, only this Digital Input will be in Error status and will not interfere with the Ready to Start status of the Whole Package.

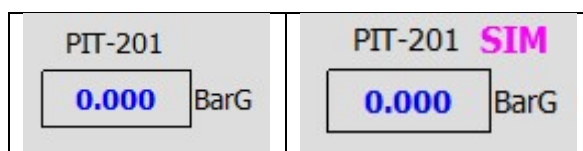
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 34 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

8.5 Analog Signals

Process Analog Signals (Instruments and calculated values) in the same way as all the read-only values, are presented with text is in colour RGB(0,0,255).

Block Icon

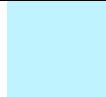
The block icon of an Analog Signal is standard in the project. For simplicity, the block icon includes just the scaled value inside a rectangle, a status text indicating when Signal is in Simulation mode and Tag name & units that will be dynamically showed/hidden by a button for all the equipment.



Faceplate

The faceplate is standard for all Analog signals and contains several statuses with different settings, features and indications useful for operation. For all the tabs, always the tag name and type are shown on the header.

Colour coding is applied also to the faceplate, for buttons and indicators as follows:

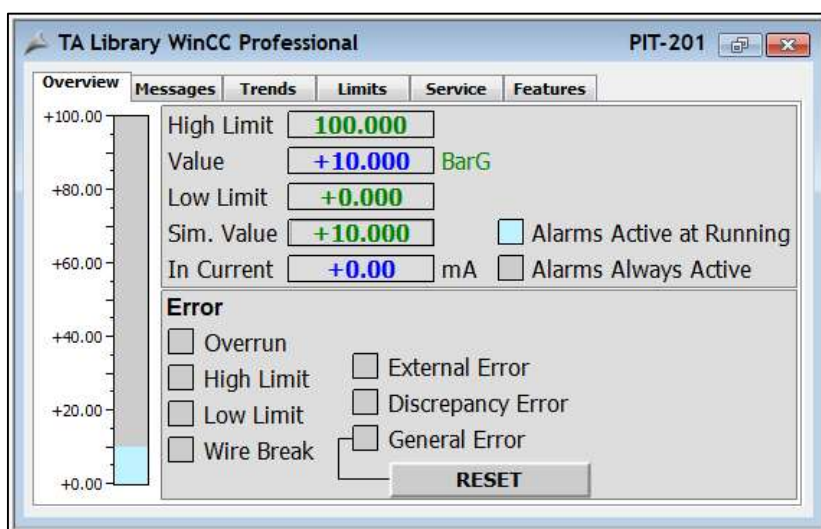
Status	RGB Code	Sample
Deactivated Status	204,204,204	
Active 1	190,243,255	
Active 2	255,255,255	
Error indication	255,170,170	

Active 1 is used when the normal/ideal mode is active (Like Monitoring ON, etc). Active 2 is used for the rest of statuses.

Simulation mode, Alarm and Trip follow their respective colour coding.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 35 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Overview tab



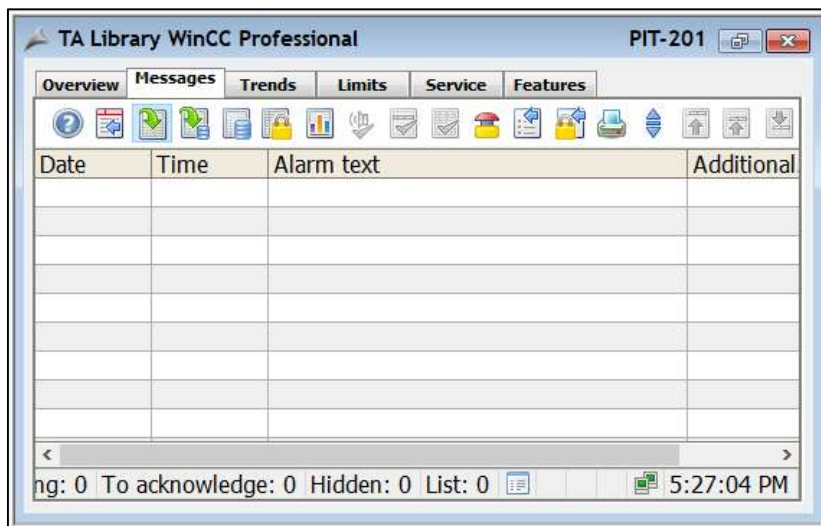
On the Overview tab, several statuses and parameters are presented as:

- 1) Graphic representation in a bar of the Analog signal value with the High and Low Limits of its scale.
- 2) Here are presented:
 - High Limit for Scaling. What 20 milliamperes are equal to in engineering units.
 - Scaled value of the Analog Signal. Engineering units are presented next.
 - Low Limit for Scaling. What 4 milliamperes are equal to in engineering units.
 - Simulation Value. When Simulation Mode is activated, this is the value the control block of the Analog signal will pass to the control logic of the System. Alarms and Trips will use this value as well.
 - Calculated value that Analog Input Module is reading in milliamperes.
- 3) Analog Signals have more statuses than any other equipment in a compressor system. Below these statuses are described:
 - Overrun: When the current in milliamperes read by the Analog Inputs Module exceeds the maximum limit allowed. When this happens, this error will generate a "Discrepancy Error" as well, being this a combined indicator related to any problem with the physical signal.
 - High Limit: When the current in milliamperes read by the Analog Inputs Module is above the normal operational range of 4 to 20 mA but has not reached the maximum high limit yet. This indication is just a warning and doesn't cause any error.
 - Low Limit: When the current in milliamperes read by the Analog Inputs Module is under the normal operational range of 4 to 20 mA but has not reached the maximum low limit yet. This indication is just a warning and doesn't cause any error.

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 36 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

- Wire Break: When the current in milliamperes read by the Analog Inputs Module is under the maximum low limit allowed (generally caused by a wire broken). When this happens, this error will generate a “Discrepancy Error” as well, being this a combined indicator related to any problem with the physical signal.
 - Errors: Here are presented all the errors an Analog signal could have:
 - “External Error” is as its name says, indicates when any external error that is related to the analog signal occurs.
 - “Discrepancy Error” indicates when a wiring/instrument related issue is present. This error combines “Overrun” and “Wire Break” status described previously.
 - “General Error” lights-up when there’s present any “External Error” or “Discrepancy Error”. There’s a reset button below to clear any error/trip/alarm related to the analog signal.
- 4) Indicators about when the Alarm and Trip limits associated with the analog signal are active. “Alarms Active at Running” lights up when Alarms/Trips are configured to be actively monitored only when the running condition of the compressor or a particular signal is received (for more details refer to the Service Tab of this faceplate). “Always Active” on the other hand, indicates Alarms/Trips are actively monitored all the time, even when compressor is not running neither starting up

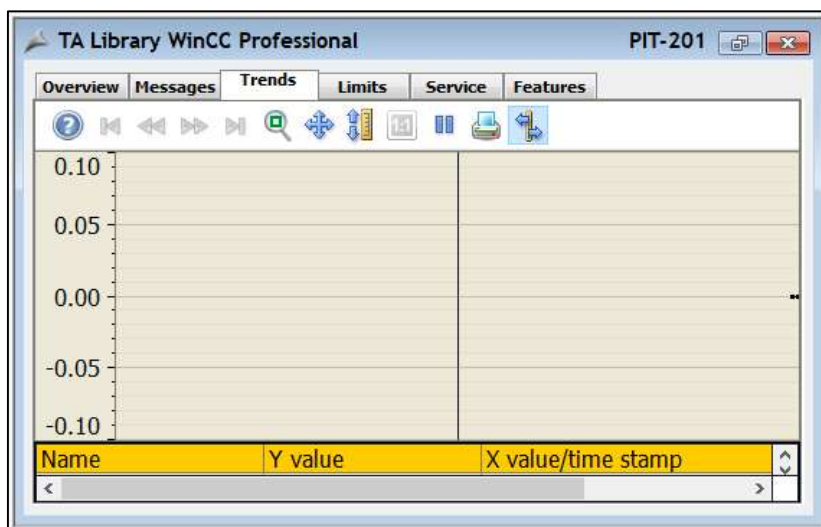
Message Tab



On “Messages” tab, user can find an alarm window that displays only the messages related to the Analog Signal who the faceplate belongs to. Time when the message was triggered, status of the alarm and description are displayed. The Background coding used for the general Alarm screen is applied as well for specific alarm messages.

Trends Tab

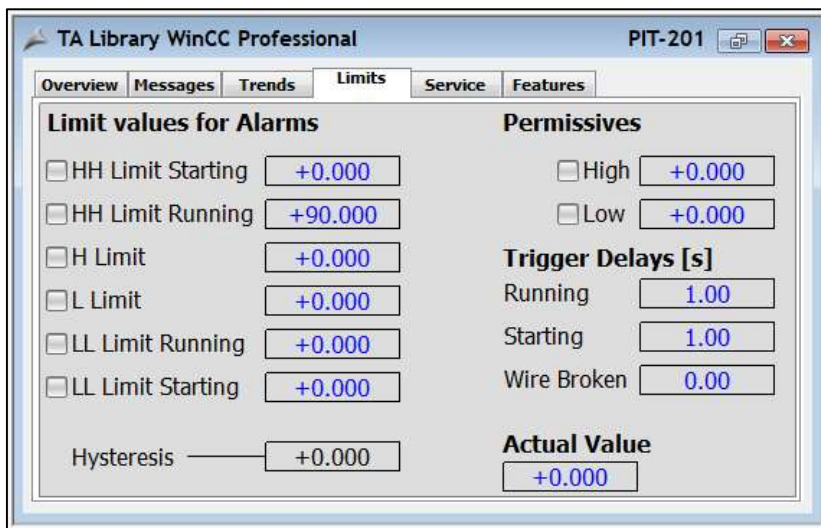
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 37 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1



On the Trends tab, a trend window including several important values related to the analog signal are being registered by the trending tool.

These values are being captured as long as the faceplate remains open. If the faceplate is closed and then re-opened, the recording starts all over again

Limits Tab



Limit values for Alarms		Permissives	
☐ HH Limit Starting	+0.000	☐ High	+0.000
☐ HH Limit Running	+90.000	☐ Low	+0.000
☐ H Limit	+0.000	Trigger Delays [s]	
☐ L Limit	+0.000	Running	1.00
☐ LL Limit Running	+0.000	Starting	1.00
☐ LL Limit Starting	+0.000	Wire Broken	0.00
Hysteresis ——— +0.000		Actual Value	
		+0.000	

On “Limits” tab, several statuses and features are presented as:

- 1) Limit Values for Alarms and Trips. Below there is a detailed description of each setting:
 - High-High Limit when Starting: This is the High Trip Setpoint used when Compressor is executing the start-up sequence if Enabled (Start-up are all the

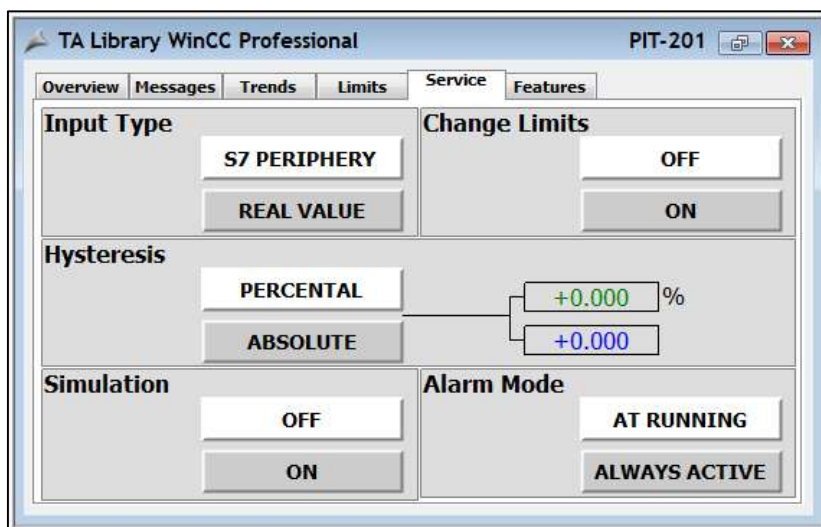
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 38 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

steps in the sequence from the moment the start command is received until Idling Mode is done. When Idling Mode is done, compressor is Running and not starting-up anymore). Checkbox indicates this setpoint is enabled and used for start-up, when is unchecked, compressor will use the “High-High Limit Running” for all the cases if it’s enabled.

- High-High Limit when Running: This is the High Trip Setpoint used when Compressor is fully running and/or all the time, depending on “Alarms Mode” setting (Refer to Service Tab for more information). Checkbox indicates this setpoint is enabled.
 - High Limit: This is the High warning/alarm Setpoint. Checkbox indicates this setpoint is enabled. When alarms start to be monitored and generated depends on “Alarms Mode” setting (Refer to Service Tab for more information).
 - Low Limit: This is the Low warning/alarm Setpoint. Checkbox indicates this setpoint is enabled. When alarms start to be monitored and generated depends on “Alarms Mode” setting (Refer to Service Tab for more information).
 - Low-Low Limit when Running: This is the Low Trip Setpoint used when Compressor is fully running and/or all the time, depending on “Alarms Mode” setting (Refer to Service Tab for more information). Checkbox indicates this setpoint is enabled.
 - Low-Low Limit when Starting: This is the Low Trip Setpoint used when Compressor is executing the start-up sequence if enabled (Start-up are all the steps in the sequence from the moment the start command is received until Idling Mode is done. When Idling Mode is done, compressor is Running and not starting-up anymore). Checkbox indicates this setpoint is enabled and used for start-up, when is unchecked, compressor will use the “Low-Low Limit Running” for all the cases if it’s enabled.
- 2) Hysteresis value used for the alarms. For more information about if it’s in absolute or percentage refer to Service Tab.
 - 3) High and Low Permissive values for analog signal to give ready status to the compressor: Enabled or disabled by the checkboxes and adjusted by their respective setpoints.
 - 4) Trigger Delays for Starting and Running Trip generation; and Trigger Delay for Wire/sensor failure generation.
 - 5) Current reading of the Analog signal (after scaling if the signal is scaled).

Service Tab

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 39 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

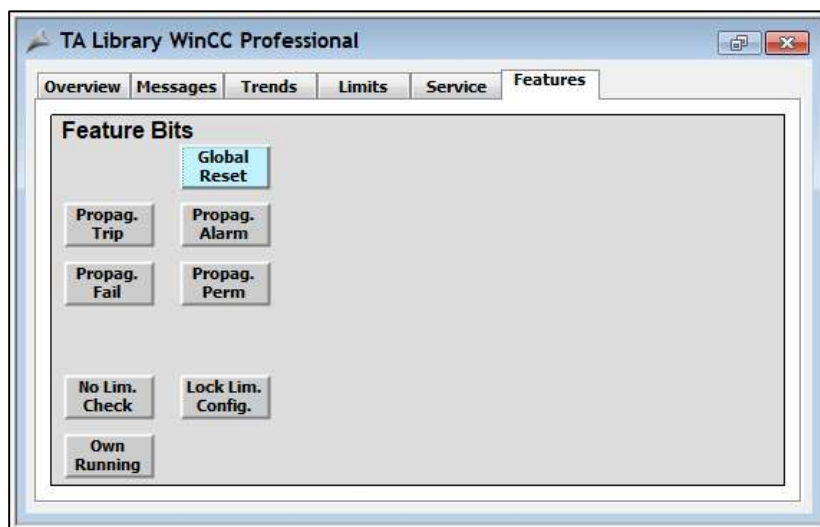


On “Service” tab, several statuses and features are presented as:

- 1) Input Type: “S7 Periphery” is chosen when the Control Block in the PLC Program is the one will scale the raw data that is coming from an Analog Inputs channel. “Real Value” must be chosen when the signal coming to the Analog Signal Control Block is scaled already and is a 32-bit floating point tag.
- 2) Change Limits: To be able to activate/deactivate Alarms and/or Trips and adjust their setpoints, must click “On” to enable those objects in the Faceplate. Once changes are made, is a good practice to click “Off” to disable changes on those objects.
- 3) If Hysteresis is used for Alarm/Trip calculations, it can be adjusted and Percentage or Absolute in Engineering Units on this area. The Hysteresis value field is enabled for editing when “Enable Change Limits” is ON.
- 4) Simulation Mode settings. Simulation mode can be activated here. In Simulation mode, the real scaled reading of the Analog signal is not used for Control neither for Alarm/Trip generation. Instead, the Simulated Value that is now visible on “Overview Tab” is used. This mode **MUST BE USED ONLY** during commissioning and/or troubleshooting **WITH TA PERSONNEL APPROVAL** and not during normal operation.
- 5) Alarms Mode. This setting defines when Alarms and Trips are generated.
 - Always Active: Alarms/Trips are actively monitored all the time, even when compressor is not running nor starting up. Setpoint used for Trip generation depends on if Startup Setpoints were enabled and if the compressor is starting up or not.
 - Active at Running: Alarms/Trips are actively monitored only when the running condition of the compressor or a particular signal is received (for more details refer to the Feature Bits of this faceplate).

Features Tab

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 40 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1



These are the main features that TA Library has. Based on the requirements of every project, these features are enabled/disabled. Light Blue background means is enabled, grey background means the opposite. An Analog Signal has the following features:

- **Global Reset:** When enabled, Analog Signal will accept global requests sent from the Compressor main control block to reset all Alarm, Trips and Failures of the equipment of the System. When disabled, reset commands can be done by Analog Signal faceplate exclusively.
- **Propagate Trip:** When enabled, if Analog Signal has been configured as a Trip source, any error status of Analog Signal (High-High Trip, Low-Low Trip, External Fault, Interlock) will send a Global Trip to the system, tripping the whole package. When disabled, only this Analog Signal will be in Error status if any error occurs (High-High Trip, Low-Low Trip, External Fault, Interlock).
- **Propagate Alarm:** When enabled, if the Analog Signal has been configured as an Alarm source, any Alarm status (High Alarm, Low Alarm, External Fault, Interlock) of Analog Signal will send a Global Alarm to the System, though will not trip the whole package. When disabled, no alarm status will be sent to Main control block, only this Analog Signal will be in Error status if any error occurs (High Alarm, Low Alarm, External Fault, Interlock).
- **Propagate Failure:** When enabled, if Analog Signal has been configured as a Trip source, Overrun/Wire Broken status will send a Global Failure to the System, tripping the whole package. When disabled, only this Analog Signal will be in Error status if is detected Overrun/Wire Broken Status.
- **Propagate Permissive:** When enabled, any error present in the Analog Signal (Overrun, Wire Broken, External Fault, Interlock, High-High Trip, Low-low Trip) will send a No-Ready status to the System and the package will not be able to Start until the error is cleared. When disabled, only this Analog Signal will be in Error status and will not interfere with the Ready to Start status of the Whole Package.
- **No Check of Scaling Limits:** When enabled, Analog signal after scaling is neither capped by the High Limit nor the Lower Limit for scaling and the scaled value will

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 41 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

be shown in the HMI application and used in the PLC logic. When disabled, any scaled value of the analog signal is capped by the scaling limits. (Example: A signal that is scaled between 0-18 Bar can show 18.5 Bar and above if this feature is enabled)

- Lock Limits Configuration: When enabled, Alarm and Trip related settings cannot be changed nor activated/deactivated.
- Own Running Signal for Active at Running mode: When enabled, and if the “Alarm Mode” used is “Active at Running”, the Analog Signal will actively monitor its Alarm/Trip settings when its Specific Running indicator digital input in its Control Block in the PLC is received and won’t use the General Running Status Indicator coming from the System Main Control Block. This feature is particularly useful when an analog signal must monitor another specific equipment running. When disabled, and if the “Alarm Mode” used is “Active at Running” the Analog Signal will actively monitor its Alarm/Trip settings when receives the General Running Status Indicator coming from the System Main Control Block.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 42 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

8.6 PID Controllers

BLOCK ICON

They are represented as blocks with their Process Value, Setpoint, Control Value and operative mode.

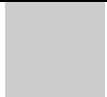
<table> <tr><td colspan="2">SUCT PRESS</td></tr> <tr><td>+0.000</td><td>BarG</td></tr> <tr><td>+0.000</td><td>BarG</td></tr> <tr><td>+0.000</td><td>%</td></tr> <tr><td colspan="2">MANUAL</td></tr> </table>	SUCT PRESS		+0.000	BarG	+0.000	BarG	+0.000	%	MANUAL		<table> <tr><td colspan="2">SUCT PRESS</td></tr> <tr><td>+0.000</td><td>BarG</td></tr> <tr><td>+0.000</td><td>BarG</td></tr> <tr><td>+0.000</td><td>%</td></tr> <tr><td colspan="2">AUTO</td></tr> </table>	SUCT PRESS		+0.000	BarG	+0.000	BarG	+0.000	%	AUTO	
SUCT PRESS																					
+0.000	BarG																				
+0.000	BarG																				
+0.000	%																				
MANUAL																					
SUCT PRESS																					
+0.000	BarG																				
+0.000	BarG																				
+0.000	%																				
AUTO																					

Please note Automatic mode that will be the standard mode for all the components is represented on a different colour, this to make easier and faster for operator to check pages with several actuators. Colour RGB(0,0,255)

FACEPLATE

The faceplate is standard for all PID Controllers and contains several statuses with different features and indications useful for operation. For all the tabs, always the tag name and type are shown on the header.

Colour coding is applied also to the faceplate, for buttons and indicators as follows:

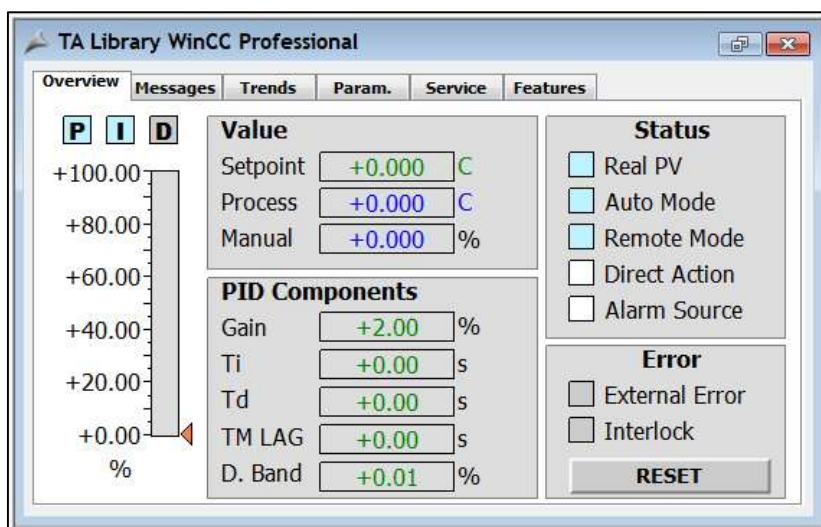
Status	RGB Code	Sample
Deactivated Status	204,204,204	
Active 1	190,243,255	
Active 2	255,255,255	
Error indication	255,170,170	

Active 1 is used when the normal/ideal mode is active (Like Remote Mode, Proportional Component active, etc). Active 2 is used for the rest of statuses.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 43 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

Simulation mode follows its colour coding.

Overview Tab



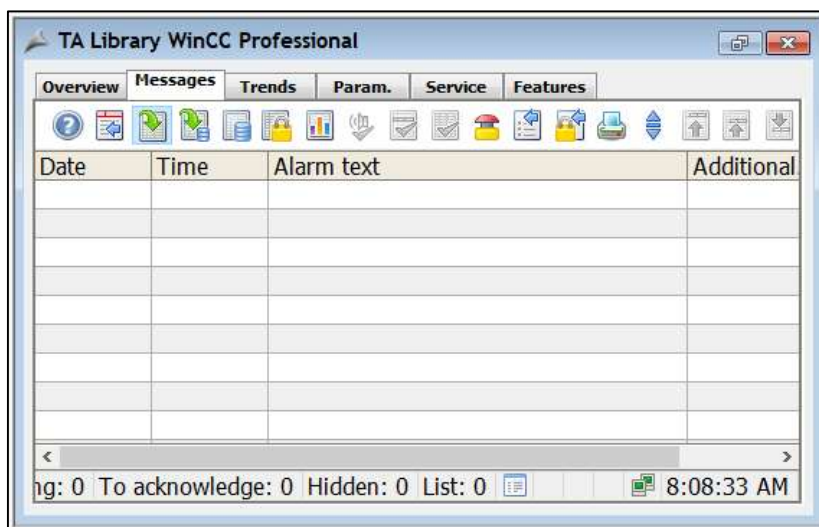
On the Overview tab, several statuses are presented as:

- 1) Graphical Indicators such as:
 - P, I, D indication boxes that light up when the respective component in the PID Controller is enabled.
 - Graphic Bar for the Process Value (PV) of the PID Controller.
- 2) Important Setpoints and commands such as:
 - Setpoint: The Target value PID Controller is trying to take the Process Value to.
 - Process Value: The reading of the parameter PID Controller is trying to control.
 - Manual Value: Is the PID Controller Output sent through its Manipulated Value (LMN) towards the Actuator when it's in Manual Mode. When PID Controller is in Override Mode, the value PLC Logic is sending to LMN is presented here.
- 3) PID Features and Indicators:
 - Process Value Indicator: Text dynamically changes showing whether the Process value is connected to the PID Controller as a scaled-real value or as a Peripheral value that is scaled in the block.
 - Mode of Operation indicator: Changes its text dynamically:
 - Auto Mode: When Control Value (output of the PID Controller) comes from PID Calculations.
 - Manual Value: Lights up when Manual Mode is activated.
 - Override Indicator: When this indicator is visible, PID Controller has been sent to Override Mode, which means Manual Mode is activated and PID Controller is controlled by PLC logic that needs it to be at a certain state in that stage of the logic.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 44 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

- Alarm Type: “Alarm Source” indicates, when PID Controller has any active Error (External Fault, Interlock Error) it will make PID Controller be in Alarm Status. “Trip Source” will take PID Controller to Trip Status. Note this status affects only this PID Controller unless the respective feature Bit is enabled to forward the status to the Compressor System.
- Remote/Local Mode: Showing the corresponding text that is changed dynamically by the mode PID Controller is at.
- PID direct/inverse action: Showing the corresponding text that is changed dynamically by the mode PID Controller is at.
- External Error: Lights up when any external error that is related to the PID Controller occurs. It's detected by the PID Controller Control Block. (This feature is not used in this project).
- Interlock: Lights up if PID Controller control block's input for Interlock supplies that signal. (Not used in this project).

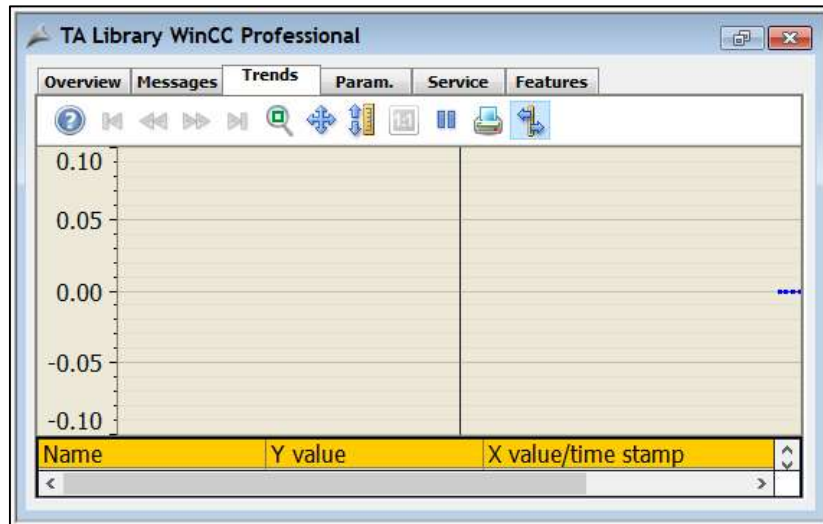
Messages Tab



On “Messages” tab, user can find an alarm window that displays only the messages related to the PID Controller who the faceplate belongs to. Time when the message was triggered, status of the alarm and description are displayed. The Background coding used for the general Alarm screen is applied as well for specific alarm messages.

Trends Tab

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 45 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1



On the Trends tab, a trend window including several important values related to the PID Controller are being registered by the trending tool.

These values are being captured as long as the faceplate remains open. If the faceplate is closed and then re-opened, the recording starts all over again.

Parameters Tab



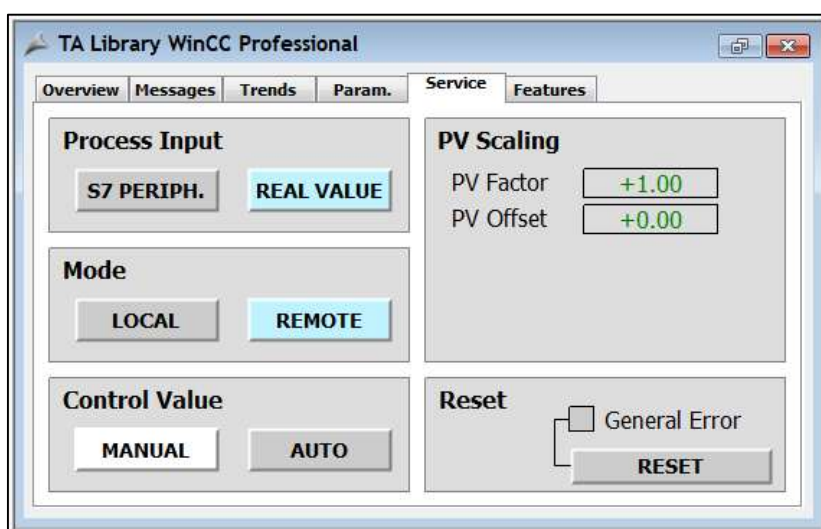
On the Parameter tab, several statuses are presented as:

- 1) Proportional component of the PID Controller. Checkbox to enable it and Gain value.
- 2) Integral component of the PID Controller. Checkbox to enable it and Ti value. Checkbox for the I-Action Hold, I-Action Initialization and Initialization Value.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 46 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

- 3) Derivative component of the PID Controller. Checkbox to enable it, Td Value and TM Lag Value.
- 4) Manipulated Value Scaling: This Factor and Offset can be used to scale the Control Value (Manipulated Value) for its use in the PID Controller Block.
- 5) Manipulated Value Limits: Manipulated Value is limited by these setpoints. They are the maximum and minimum output at LMN the PID Controller will provide.
- 6) Manipulated Value warning settings. When process value limits are enabled and it is above/below the high/low setpoints, warning indications are generated. Note these warnings will not affect Trip/Alarm status of the compressor system, are just for information purposes.

Service Tab



On the Service tab, several statuses are presented as:

- 1) Process Value Source: When is in S7 Periphery, the source of this value comes directly from the raw data from an Analog Input Channel. When it is set in Extern, there's another block (normally an Analog Signal Block) that does the scaling and its data is connected to the PID Controller.
- 2) Local and Remote mode for the source of command and parameters. PID Control in this project is all the time in Remote Mode.
- 3) Control Value: In Manual Mode, the operator by the faceplate in the overview tab (and when Override mode is not activated) can set the value to be sent by PID Controller LMN output. In Process Mode, the PID Algorithm is the one calculating the value for LMN output. Manual mode IS NOT THE NORMAL MODE OF OPERATION and must not be enabled without prior approval and/or presence of TA personnel.
- 4) Periphery Scaling: When Process Value is read as raw data directly from the Analog Signals Module, this Factor and Offset can be used to scale the Process Value for its use in the PID Controller Block.

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 47 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

- 5) Reset Button to clear any Error related to the PID Controller. General Error Indicator is displayed as well.

Features Tab



- 6) These are the main features that TA Library has. Based on the requirements of every project, these features are enabled/disabled. Light blue background means is enabled, grey background means the opposite. A PID Controller has the following features:
- **Global Auto:** When enabled, PID Controller will accept global requests sent from the System main control block to switch to Auto mode all the equipment of the system. When disabled, switching to auto mode can be done by PID Controller faceplate exclusively.
 - **Global Reset:** When enabled, PID Controller will accept global requests sent from the System main control block to reset all Alarm, Trips and Failures of the equipment of the system. When disabled, reset commands can be done by PID Controller faceplate exclusively.
 - **Propagate Trip:** When enabled, if PID Controller has been configured as a Trip source, any error status of PID Controller (except feedback error) will send a Global Trip to the System, tripping the whole package. When disabled, only this PID Controller will be in Error status if any error occurs (External Fault, Interlock).
 - **Propagate Alarm:** When enabled, if the PID Controller has been configured as an Alarm source, any Alarm status of PID Controller will send a Global Alarm to the System, though will not trip the whole package. When disabled, no alarm status will be sent to System main control block, only this PID Controller will be in Error status if any error occurs (External Fault, Interlock).
 - **Propagate Permissive:** When enabled, any error present in the PID Controller (External Fault, Interlock) will send a No-Ready status to the System and the package will not be able to Start until the error is cleared. When disabled, only this

 ManaEnergy Managem Group	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 48 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

PID Controller will be in Error status and will not interfere with the Ready to Start status of the whole Package.

- Propagate Auto Mode: When enabled, if the PID Controller is in Manual Mode, it will send a No-Ready status to the Compressor system and Compressor package will not be able to Start until it is switched to Auto Mode and a Reset Alarm command is executed. When disabled, Manual/Auto mode will not interfere with the Ready to Start status of the Package. Please note Override Mode is the normal status of the PID Controller when the compressor is not running, since in this mode, the Logic is the one commanding the PID Controller and its output.
- Follow Auto/Manual Mode from Actuator: When Enabled, PID Controller will change its Manual/Auto Mode to match the mode set in its actuator (Slide Valve or Control Valve depending on which PID Controller in the project we speak about). When Disabled, PID Controller Auto/Manual Mode will depend exclusively on the Mode set using the faceplate and if Override mode is requested by Control Logic. (This feature is disabled by Default in the present Project)

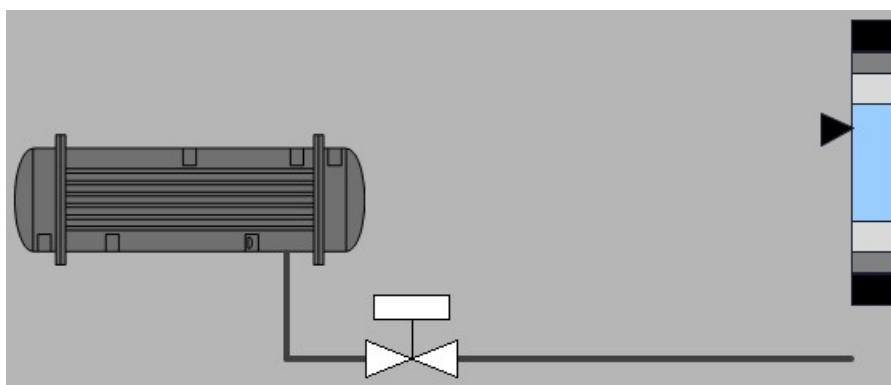
	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 49 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1

8.7 Interlocks

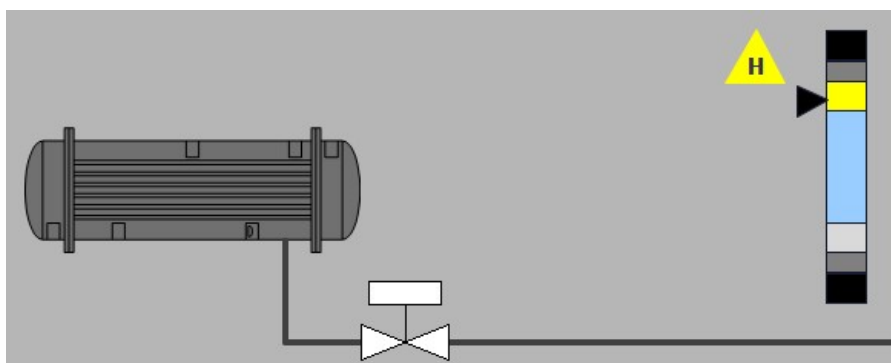
Interlocks are conditions that when met, disable/block/lock actuators and controls to a specific state. Based on recommendations and guidelines consulted, they will be animated and indicated on the respective process screen following the colours defined previously in “Alarm Indications” part of this document.

Next there’s an example of how Interlocks will be represented on the system:

On/Off Valve opened:



On/Off Valve opened, alarm range reached for a given Analog value:



On/Off Valve Closed because Analog Value has reached Interlock range. Visual indication following colour code is visible, together with a line and frame indicating interlock source and actuator interlocked:

	Sound Energy MANAENERGY	Document Code 21044-AU-HMI-00002	Sheet of Sheets 50 / 50	
	Contractor ITALFLUID MOROCCO BRANCH		Date	Rev.
	Engineering ITALFLUID Geoenergy S.r.l.	Document Title HMI PHILOSOPHY	16/01/2026	1
				

